



SOEN 6841: SOFTWARE PROJECT MANAGEMENT

TOPIC: FINANCIAL LITERACY APP

**PHASE II – FEASIBILITY STUDY, SOLUTION PROPOSAL, PROJECT PLAN, RISK
ASSESSMENT AND BUDGETING DOCUMENT**

Group 1

Angad Singh Sodhi (40325097)

Joshua Onyema (40293065)

Devshree Shah (40269569)

Devang Shah (40306683)

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FEASIBILITY STUDY AND SOFTWARE SOLUTION PROPOSAL

1.1 Objective

This feasibility study aims to assess the viability of developing a Stock Selection and Financial Literacy App that enables retail and institutional investors to make informed, data-driven, and emotion-free investment decisions. The application leverages High Quantitative Momentum (HQM) analysis to recommend stock portfolios based on user risk preferences, market data, and long-term investment strategies.

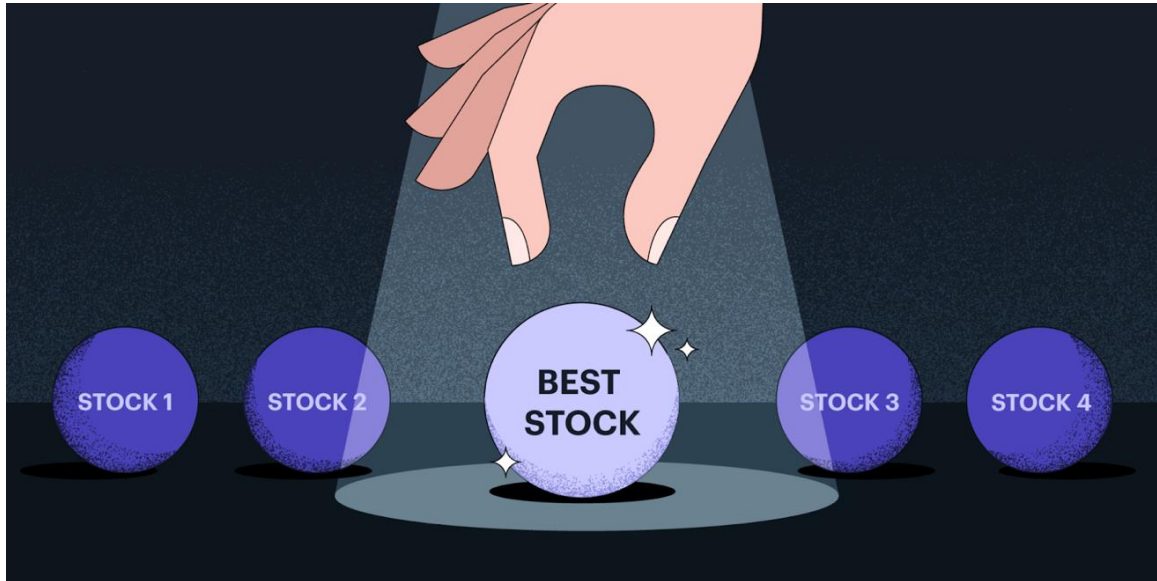


Figure 1: Image depicting how HQM works

1.2 Technical Feasibility

- i. Evaluation of the Technology Requirements for the Software Solution: The application will utilize Python as the core backend language, with data science libraries like Pandas, NumPy, and Scikit-learn for analytics and machine learning. React and Flutter will be used to develop web and mobile interfaces, respectively, ensuring cross-platform compatibility and a seamless user experience. APIs such as IEX Cloud, Alpha Vantage, and Yahoo Finance will be integrated for real-time and historical stock data. Cloud platforms like AWS or Firebase will be used for hosting, storage, and authentication. Security protocols will include end-to-end encryption, OAuth 2.0, and two-factor authentication. Scalability will be achieved through containerized deployment using Docker and Kubernetes, ensuring smooth performance even as user demand grows.
- ii. Assessment of the Feasibility of Implementing the Required Technology: All technologies considered are well-documented, widely used, and supported by active developer communities. Technical risks include potential API rate limits, data inconsistency, and integration challenges. However, alternatives such as paid API tiers and redundant services mitigate these risks. The

technology stack is also adaptable to future enhancements, including machine learning integration and predictive modelling.

iii. Risk vs. Reward Analysis: Python was chosen for its simplicity, vast libraries, and flexibility in financial modelling. Alternatives like Java or C++ offer performance benefits but are less agile in prototyping and have longer development cycles. React was selected for its performance, large ecosystem, and reusability, while Flutter enables fast mobile development with a single codebase. These trade-offs prioritize development speed and scalability without compromising performance.

1.3 Operational Feasibility

i. Analysis of the Operational Impact of the Proposed Solution on Existing Processes: The application will modernize the investment decision-making process by shifting from emotional and manual analysis to automated and data-driven workflows. Roles such as financial advisors will be supplemented, if not replaced, with intelligent automation. New workflows will include daily stock updates, portfolio monitoring, and risk tracking. The operational efficiency will improve as manual tasks are automated. Training for staff and end-users will be minimal due to the user-centric design, enhancing adoption and reducing onboarding time. Dashboards, tutorials, and in-app help will further ease the learning curve.

ii. Identification of Potential Challenges and Benefits in the Operational Context: Challenges include user resistance to automation, especially among traditional investors, the need for initial training, and infrastructure readiness. Benefits include cost reductions, increased efficiency, improved investment outcomes, and enhanced transparency.

iii. Transition Plan: Implementation will follow a phased rollout. Training modules, user guides, and in-app tutorials will facilitate smooth onboarding. Pilot testing with selected user groups will refine the solution before full-scale deployment. A dedicated support team will be available during the initial deployment phase to ensure seamless adoption and provide timely assistance. Continuous feedback loops will support iterative improvements and help maintain user engagement.

1.4 Economic Feasibility

i. Estimation of the Economic Viability of the Project: Development costs, including frontend, backend, database, and API integration, are estimated at \$25,000. Third-party API subscriptions are projected to cost approximately \$2,400/year, based on current pricing. Testing and QA, including manual, automated, and user acceptance testing, are budgeted at \$5,000. Operational infrastructure, such as cloud services and deployment, is expected to require \$3,000–5,000 initially, with ongoing maintenance costs estimated at \$5,000 annually. Long-term expenses may include feature enhancements and server scaling as the user base grows.

ii. Consideration of Resource Availability, ROI, and Cost-Benefit Analysis: Resources such as open-source libraries, skilled freelance developers, and cloud platforms (e.g., Firebase, AWS) are readily available and cost-effective. ROI is projected at ~200–250% over 3 years, with a payback period of 18–24 months based on a freemium model, premium subscriptions, and strategic partnerships with financial institutions. A detailed cost-benefit analysis shows strong user value, especially for individuals replacing traditional financial advisors. The app could result in average annual savings of \$2,000 per user, while also improving portfolio performance through data-driven, emotion-free investing strategies.

SOFTWARE SOLUTION PROPOSAL

2.1 Solution Overview

The Stock Selection and Financial Literacy App is a data-driven platform that guides users through a strategic, algorithm-backed investment process. It leverages HQM analysis, personalized risk profiling, and real-time financial data to generate stock recommendations. The system is modular and scalable, allowing for future integration with AI, additional asset classes (e.g., bonds, crypto), and direct trading via brokerage APIs. Security features include encryption, multi-factor authentication, and regulatory compliance with IIROC and CSA.

2.1.1 How It Addresses the Identified Problem:

The app addresses the widespread problem of emotionally driven, suboptimal investing by providing systematic, quantifiable investment strategies. It replaces guesswork and cognitive biases with evidence-based portfolio recommendations. Use cases show clear improvements in user confidence, investment accuracy, and long-term wealth accumulation. Compared to existing solutions (mutual funds, hedge funds, MetaTrader), it offers greater transparency, control, and accessibility.

2.2 Key Features and Functionalities

i. Detailed Listing of the Essential Features and Functionalities:

- a. Automated Stock Shortlisting: Based on HQM analysis using momentum indicators and performance metrics.
- b. Custom Risk Profiling: Users can define their risk appetite (conservative, moderate, aggressive).
- c. Real-time Data Integration: Continuously fetches and analyses stock data.
- d. Simulation Mode: Allows users to test hypothetical strategies.
- e. Customizable Dashboard: Visual tools for tracking, editing, and optimising portfolios.
- f. Education Hub: Offers interactive learning modules on investing and financial literacy.

Each feature is prioritised based on ease of use, demand, and market research, ensuring maximum value to users.

ii. Use Cases or Scenarios Illustrating How Users Will Interact with the Solution:

- a. A novice investor logs in, completes a risk assessment, and is guided through a recommended portfolio with educational tips.
- b. A hedge fund manager imports data from the app into internal systems to optimize asset allocations.
- c. A retiree uses simulation mode to plan long-term returns based on conservative strategies.
- d. An analyst uses real-time data views and insights to validate market movements and rebalance holdings.

Each use case includes user input, system processing, feedback loops, and analytics outputs to help users iterate on strategy.

2.3 Benefits and Impact

i. Clear Articulation of the Benefits that Users and Stakeholders Will Derive from the Solution:

- a. Users gain access to institutional-grade investing tools with zero prior expertise required.
- b. Financial managers can optimize time and performance via automation and analytics.
- c. Stakeholders benefit from increased adoption, brand recognition, and monetization opportunities.

Short-term benefits include time savings and improved decision quality; long-term benefits include financial literacy growth, better ROI, and higher portfolio stability.

ii. Expected Impact on the Target Audience and the Broader Domain:

For individuals, particularly retail investors, young professionals, first-time investors, and those lacking access to traditional financial advisors, the app offers a transformative experience. It empowers users with limited financial literacy to make data-driven investment decisions through an intuitive, guided platform. By removing emotional bias and simplifying stock selection, the app promotes long-term wealth-building and financial independence.

Users who previously avoided investing due to complexity or fear of making mistakes can now engage with confidence. The inclusion of educational features, simulations, and personalized insights ensures that even novice investors are equipped to navigate markets effectively.

On a broader scale, the solution supports financial inclusivity by bridging the gap between casual investors and institutional-grade strategies. It contributes to increased market participation, especially among millennials and Gen Z users who favour digital, self-directed financial tools. Furthermore, the app supports global fintech trends toward decentralised, transparent, and user-centric investing, helping to redefine standards in retail finance.

In the long term, this platform could significantly enhance financial literacy, reduce reliance on costly intermediaries, and create a more informed, empowered generation of investors.

PROJECT PLAN – WORK BREAKDOWN STRUCTURE

3.1 Objective of the Gantt Chart

The objective of this Gantt chart is to visually represent the project timeline for the development of the Financial Literacy App. It outlines key phases, tasks, and milestones from mid-Q1 to mid-Q3 of 2025, ensuring clear visibility into task durations, dependencies, and time allocations. This structured timeline aids in effective project management, resource planning, and tracking of progress aligned with strategic project goals.

3.2 Introduction Pointers for the Gantt Chart Section

- i. The Gantt chart provides a comprehensive visual timeline of the entire project lifecycle, segmented into five key phases: Planning, Design, Development, Testing, and Deployment.
- ii. It covers the project execution period from February 2025 to August 2025, beginning in mid-Q1 and concluding in mid-Q3, aligning with academic and organizational timelines.
- iii. Each phase is allocated a realistic duration based on task complexity, resource availability, and expected deliverables, ensuring effective time management.
- iv. The timeline considers task dependencies, ensuring that sequential and parallel activities are coordinated to avoid bottlenecks and ensure timely delivery.
- v. The Gantt chart serves as a planning and tracking tool for stakeholders, offering a clear overview of milestones and allowing for informed decision-making throughout the project.
- vi. By aligning tasks with project objectives and key deliverables, the chart ensures that development is systematic, efficient, and focused on achieving a successful launch of the Financial Literacy App.

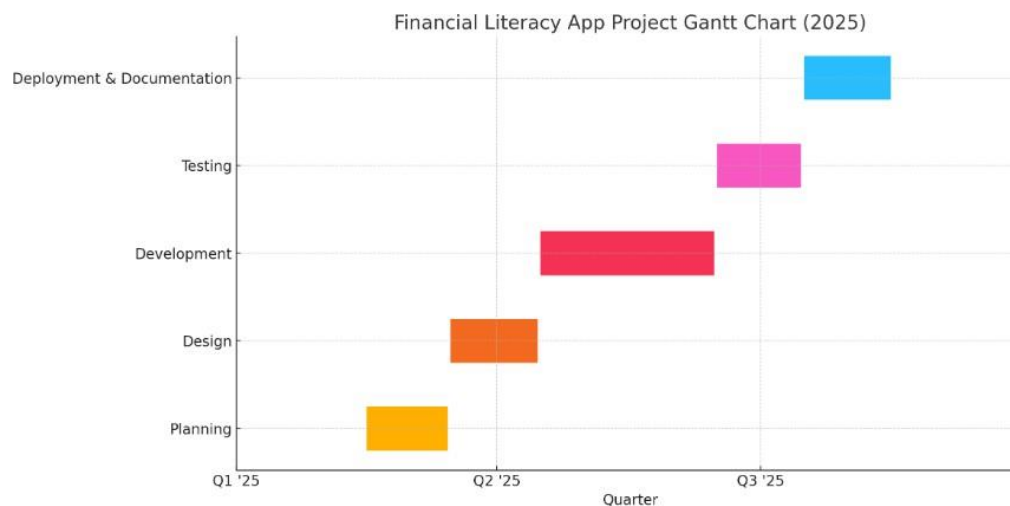


Figure 2: Financial Literacy App Project Gantt Chart (2025)

3.3 Project Phases – Tasks, Focus, and Resources

Phase	Key Tasks	Deliverables	Resources
Phase 1: Planning (4 weeks)	- Conduct stakeholder meetings - Gather and document requirements (SRS) - Perform feasibility analysis - Finalize tech stack	- Clarify project scope and objectives - Define deliverables and sign-off criteria	- Team Lead / BA: Devshree Shah - All members: Ideation & discussions - Tools: Google Docs, JIRA
Phase 2: Design & Architecture (4 weeks)	- Create system architecture (frontend + backend) - Design database schema - Build UI/UX mockups and prototypes	- Align with requirements from Phase 1 - Emphasis on user-centric design	- UI/UX: Joshua Onyema - Architect: Angad Singh Sodhi - Tools: Figma, Lucidchart
Phase 3: Development (8 weeks)	- Develop frontend (dashboard, stock analysis) - Implement backend logic (HQM analysis, APIs) - Integrate and perform unit testing	- Agile sprints, regular code reviews - Maintain modular, scalable codebase	- Frontend: Joshua Onyema - Backend: Angad Singh Sodhi - Tools: GitHub, VS Code, Postman
Phase 4: Testing (4 weeks)	- Conduct unit & integration testing - Perform User Acceptance Testing (UAT) - Bug fixes, performance testing	- Validate functionality and performance - Ensure readiness for deployment	- QA Lead: Devshree Shah - UAT: Devang Shah - Tools: JIRA, GitHub Issues, Test Logs
Phase 5: Deployment & Documentation (4 weeks)	- Setup hosting & deploy app - Create user manuals & developer guides - Conduct project closure & handover	- Smooth deployment process - User onboarding and support	- DevOps: All Members - Documentation: Devshree Shah - Tools: GitHub Pages/AWS, Word/PDF Tools



- **Criteria for Completion**

Stakeholders approve requirements, feasibility validated, tech stack selected.

- **Deliverables**

- Software Requirements Specification (SRS)
- Feasibility Report
- Tech Stack Document

- **Impact**

Provides a clear, agreed roadmap for all future phases; aligns team and stakeholder expectations.



- **Criteria for Completion**

UI/UX mockups and system architecture reviewed and approved by the team.

- **Deliverables**

- Wireframes and Mockups (Figma)
- Database Schema
- Architecture Diagrams

- **Impact**

Ensures that technical design is aligned with user needs, enabling seamless development



- **Criteria for Completion**

Frontend and backend completed, integrated, and unit tested.

- **Deliverables**

- Fully Functional Codebase (Frontend + Backend)
- API Integration Report
- Unit Test Report

- **Impact**

Demonstrates a working version of the app; ready for testing and validation.



- **Criteria for Completion**

All major bugs resolved, successful UAT conducted, app performance validated.

- **Deliverables**

- Test Case Results
- Bug Resolution Log
- UAT Feedback Summary

- **Impact**

Ensures the app is stable, user-friendly, and ready for real-world deployment.



- **Criteria for Completion**

Live deployment completed, documentation delivered, project closure approved.

- **Deliverables**

- Live App Access
- User Manual
- Developer Documentation
- Closure Report

- **Impact**

Final milestone signifying project success; ensures sustainability, user onboarding, and future maintenance support.

3.4 Deliverable Descriptions

Deliverable	What It Includes	Delivery Format
Software Requirements Specification (SRS)	Detailed functional and non-functional requirements, use cases, and system features	PDF Document (Google Docs / Word)
UI/UX Designs	Wireframes, mockups, user flow diagrams designed for optimal user experience	Figma Prototypes / PDF Export
System Architecture Document	High-level system components, data flow diagrams, backend/frontend architecture	Lucidchart / PDF Diagrams
Codebase (Frontend + Backend)	Source code files, APIs, integration scripts with version control	GitHub Repository (Tagged Release)
Test Reports	Test cases executed, results, bug logs, UAT feedback	Excel Sheets / JIRA Export
User Manual	Step-by-step guide for end users, app features, troubleshooting tips	PDF Manual
Developer Documentation	Technical details for future maintenance: API references, setup instructions, deployment guide	Markdown Docs / PDF

Project Closure Report	Summary of project outcomes, challenges, achievements, and future recommendations	PDF Document
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3.5 Task Management and Tracking

To ensure efficient task delegation, progress tracking, and milestone alignment, we implemented a structured Task Tracker using GitHub Projects. This visual and interactive board enables the project team to manage deliverables effectively across all phases—from design to deployment.

i. The task tracker is segmented into key workflow stages:



providing real-time visibility into the status of each task.

ii. Purpose and Benefits:

- Centralized Tracking:** All project tasks are documented as issues, each assigned with effort estimates and owners, ensuring accountability.
- Workflow Clarity:** Status columns help monitor progress, detect bottlenecks, and facilitate timely review and feedback.
- Enhanced Collaboration:** Team members can comment, update, and resolve issues collaboratively within GitHub, maintaining transparency.
- Milestone Alignment:** Each task is tagged with labels and milestones (e.g., Testing Completion, Deployment), ensuring deliverables align with project goals and timelines.

iii. Key Features Implemented:

- Effort Estimation:** Tasks are estimated in hours to help with time management and workload distribution.
- Labels:** Used for quick identification of task type (e.g., Design, QA, UAT, Documentation).
- Assignees:** Mapped responsibilities.
- Dynamic Updates:** Tasks dynamically move across stages as progress is made, from initial planning to final delivery.

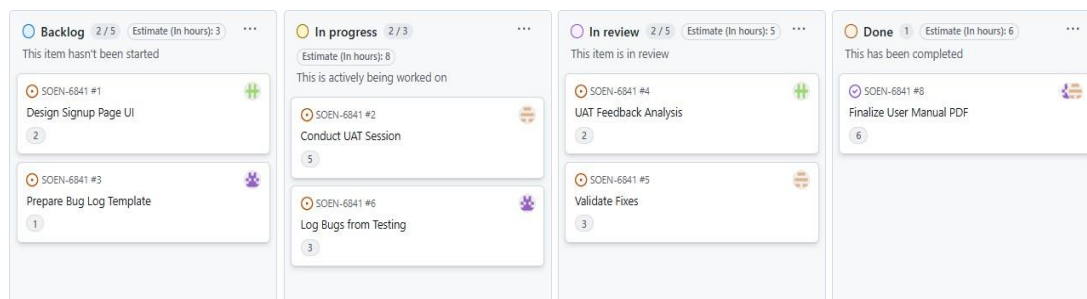


Figure 3: Kanban View

Financial Literacy App - Task Tracker [Increased items preview](#) [Feedback](#)

[Backlog](#)
[Priority board](#)
[Team items](#)
[Roadmap](#)
[In review](#)
[My items](#)
[View 7](#)
[View 8](#)
[+ New view](#)

Filter by keyword or by field

	Title	Status	Estimate (In hours)	Assignees	Labels
1	Design Signup Page UI #1	Backlog	2	sodhi-007	Backlog Design
2	Conduct UAT Session #2	In progress	5	devang-1910	Testing UAT
3	Prepare Bug Log Template #3	Backlog	1	shahdevshree	Backlog QA Testing
4	UAT Feedback Analysis #4	In review	2	sodhi-007	UAT
5	Validate Fixes #5	In review	3	devang-1910	QA
6	Log Bugs from Testing #6	In progress	3	shahdevshree	QA Testing
7	Finalize User Manual PDF #8	Done	6	devang-1910, shahdevshree, ...	documentation

+ You can use **(Control + Space)** to add an item

Figure 4: Tabular View

iv. Outcome:

This organized approach enhances project visibility, supports agile workflows, and contributes to successful and timely project completion. The GitHub task board serves as a real-time, living document of progress and collaboration.

RISK ASSESSMENT AND BUDGETING DOCUMENT

4.1 Risk Assessment and Mitigation

4.1.1 Risk Identification

i. Technical Risks

a. Risk: System Downtime or Application Failure: There could be unexpected system downtimes, server issues, or bugs in the application, causing temporary or prolonged disruption to service. This could lead to loss of user trust or data inaccessibility. This risk is particularly critical as it can significantly impact user experience and damage the platform's reputation.

b. Risk: Inaccurate Stock Recommendations due to Data Errors: The application relies on continuous data streams for stock analysis, including market data, stock performance, and risk factors. Any inconsistency in this data or failure to update it could lead to inaccurate recommendations. If the data is flawed or outdated, it could negatively affect the financial outcomes for investors and erode confidence in the platform.

ii. Operational Risks

a. Risk: Insufficient Data Quality or Data Integrity Issues: The stock selection algorithm depends heavily on accurate, timely, and high-quality financial data. Any failure in data collection, processing, or storage could compromise the integrity of the app's recommendations. The accuracy of stock recommendations is directly tied to the data quality. Operational failures could result in suboptimal investment strategies.

b. Risk: Scalability of Infrastructure: As user adoption grows, the system's infrastructure may be tested for its ability to handle large volumes of simultaneous requests or large datasets. Failure to scale properly could lead to performance bottlenecks or slower response times. Performance issues could severely hinder the user experience, especially for institutional investors who expect near-instantaneous analysis.

iii. Financial Risks

a. Risk: Budget Overruns or Insufficient Funding: There is a risk that the project may face higher-than-expected development costs, which could impact the timeline or features of the application. Budget overruns can delay the project and potentially reduce the quality or scope of features if the funding isn't properly managed.

b. Risk: Regulatory Compliance Costs: Changes in regulations or additional compliance requirements (such as data protection laws or financial regulations) may increase the development and operational costs of the application. Non-compliance or unexpected regulatory requirements could lead to fines or forced changes in the platform's features, which would affect delivery and operational costs.

iv. Market Risks

- a. Risk: Market Acceptance and Adoption: Despite the technological and financial advantages, users may be hesitant to adopt a new investment platform, especially retail investors unfamiliar with quantitative models. Low adoption or market resistance could undermine the success of the application and its growth potential.
- b. Risk: Competitive Landscape: Competitors in the algorithmic investment space (e.g., MetaTrader 5, hedge funds, mutual funds) might offer similar solutions with superior brand recognition, user base, or features. Market share could be captured by more established competitors, affecting both revenue and long-term viability.

v. Environmental Risks

- a. Risk: Changes in Economic Conditions or Financial Markets: The stock market is inherently volatile, and sudden shifts in economic conditions (e.g., recession, market crash) can alter investment strategies and the effectiveness of the algorithms used in stock selection. These market fluctuations can undermine the application's recommendations and financial models, potentially leading to losses for users.

4.1.2 Categorization of Risks

To effectively manage and address the uncertainties associated with the project, risks have been classified into five main categories based on their source and potential impact:

i. Technical Risks

These risks originate from the technologies, tools, and systems used in developing and deploying the application. System Downtime or Application Failure: Unexpected crashes, outages, or unresponsiveness of the application during use. Inaccurate Stock Recommendations: Errors or inconsistencies in data processing or model logic leading to flawed investment advice.

ii. Operational Risks

These are risks related to the day-to-day operation, data quality, and system scalability. Insufficient Data Quality or Integrity Issues: Poor data hygiene could compromise algorithm performance and user trust. Scalability of Infrastructure: Inability of the platform to accommodate increasing traffic or data loads without degradation in performance.

iii. Financial Risks

These risks pertain to project budgeting, funding, and unforeseen regulatory expenditures. Budget Overruns or Funding Gaps: Development and operational expenses exceeding projections. Regulatory Compliance Costs: Increased costs due to changing financial and data regulations or legal obligations.

iv. Market Risks

These risks involve user adoption, competition, and product relevance. Limited Market Acceptance and Adoption: Hesitance or resistance from the target audience to embrace a new platform. Competitive Landscape: Threat from more established competitors with larger user bases or more robust features.

v. Environmental Risks

Risks arising from broader economic or market forces that are beyond the project's direct control. Adverse Economic Conditions: Recession, inflation, or market volatility that could impact user behaviour and investment strategies.

4.1.3 Risk Impact Analysis

A detailed assessment of the potential consequences of each identified risk is provided below, evaluating both the severity of the impact and relevant performance metrics.

i. Individual Risk Impact Assessments

a. System Downtime or Application Failure

Impact Level: Severe

Description: Unexpected downtime or application crashes may disrupt user activities, cause data loss, and severely affect brand trust.

Consequences: Reduced platform reliability and negative user perception.

Key Metrics:

- Customer churn rate
- Downtime duration reports
- Number of unresolved support tickets
- Decrease in active daily users

b. Inaccurate Stock Recommendations

Impact Level: Very High

Description: Errors in recommendation algorithms due to flawed data or miscalculations can result in poor investment decisions for users.

Consequences: Loss of user trust, potential legal liability, negative reviews.

Key Metrics:

- Accuracy rate of predicted returns
- Frequency of data anomalies
- Volume of user complaints linked to recommendations
- User retention after financial losses

c. Insufficient Data Quality or Integrity Issues

Impact Level: High

Description: Poorly sourced or unclean data can impair model performance and produce misleading outputs.

Consequences: Decreased user satisfaction, ineffective algorithm decisions.

Key Metrics:

- Data validation failure rate
- Incidents of inconsistent or missing data
- Accuracy degradation over time

d. Scalability Limitations

Impact Level: Moderate to High

Description: A surge in user numbers or data demands may overwhelm system infrastructure.

Consequences: Lagging, timeouts, and degraded user experience under load.

Key Metrics:

- Response time under peak loads
- Server CPU/memory utilization metrics
- User feedback during peak hours

e. Budget Overruns or Funding Shortfalls

Impact Level: Moderate

Description: Underestimation of project costs or funding gaps could delay timelines or reduce scope.

Consequences: Postponed milestones, incomplete features, or quality compromise.

Key Metrics:

- Actual vs. projected expenditure
- Burn rate deviation
- Delays in deliverables

f. Regulatory Compliance Costs

Impact Level: Moderate

Description: New or unexpected compliance requirements (e.g., data protection laws) may require redesign or additional resources.

Consequences: Additional legal consultation, platform changes, or launch delays.

Key Metrics:

- Compliance audit reports
- Legal fees incurred
- Change requests tied to compliance

g. Market Acceptance and Adoption

Impact Level: High

Description: Target users may be reluctant to transition from traditional investment methods to a digital, algorithm-based solution.

Consequences: Low user acquisition and subscription rates.

Key Metrics:

- Conversion rate from trial to subscription
- Monthly active users
- Net Promoter Score (NPS)

h. Competitive Landscape

Impact Level: High

Description: Larger or better-known platforms may introduce similar features or aggressively capture market share.

Consequences: Loss of potential users, need to differentiate further.

Key Metrics:

- Market share comparisons
- Feature benchmarking analysis
- Customer switch rate

i. Economic Conditions and Market Volatility

Impact Level: High

Description: Recessions or downturns can discourage investment activities, impacting user engagement and platform utility.

Consequences: Drop in platform usage, and reduced value of recommendations.

Key Metrics:

- Platform activity correlated with market trends
- User withdrawal or inactivity rate
- Investor confidence indices

ii. Prioritization of Risks Based on Severity and Likelihood

All identified risks have been plotted using a Probability × Impact Matrix, allowing for visual prioritization and strategic planning. Risks with high probability and high impact (e.g., inaccurate recommendations, market rejection) are treated with the highest urgency. This prioritization is informed by historical data from similar fintech projects and expert judgment from domain consultants.

Risk	Likelihood	Impact	Priority	Metrics
System Downtime	High	Very High	Critical	Customer churn, negative reviews, loss of market share
Inaccurate Stock Recommendations	Medium	Very High	Critical	Financial losses for users, negative feedback
Insufficient Data Quality	High	High	High	Reduced prediction accuracy, user dissatisfaction
Scalability	Medium	High	High	Slow response times, customer complaints
Budget Overruns	Medium	Medium	Medium	Extended timelines, reduced functionality
Regulatory Compliance	Low	Medium	Medium	Increased compliance costs, delayed product release
Market Acceptance	Medium	High	High	Adoption rate, market share
Competitive Landscape	High	High	High	Customer acquisition rate, user retention
Changes in Economic Conditions	Medium	High	High	Market volatility, user losses

Figure 5: Risk Analysis table

4.2 Risk Mitigation Strategies

4.2.1 Development of Strategies to Mitigate or Minimize the Impact of Identified Risks

i. System Downtime or Application Failure

Mitigation Strategy:

Implement a proactive system maintenance schedule, cloud infrastructure with failover redundancy, automated daily backups, and 24/7 performance monitoring. This ensures service continuity and quick recovery in case of disruption.

ii. Inaccurate Stock Recommendations

Mitigation Strategy:

Utilize multiple, reputable financial data sources to ensure redundancy and accuracy. Integrate real-time data validation checks and automated monitoring systems to flag anomalies or lags in data updates.

iii. Insufficient Data Quality or Integrity Issues

Mitigation Strategy:

Collaborate with high-trust data providers. Incorporate machine learning models for real-time data cleaning and consistency checks. Establish strict data validation protocols before feeding inputs to algorithms.

iv. Scalability of Infrastructure

Mitigation Strategy:

Design a scalable system architecture using microservices and containerized deployments (e.g., Docker, Kubernetes). Perform stress/load testing periodically and enable horizontal scaling to meet demand.

v. Budget Overruns or Funding Gaps

Mitigation Strategy:

Create a comprehensive budget plan with contingency buffers. Track expenditures regularly through project management tools and reallocate resources efficiently. Secure flexible funding options if additional capital is needed.

vi. Regulatory Compliance Costs

Mitigation Strategy:

Engage compliance experts early in the development lifecycle. Monitor regional regulations proactively and design the system to be adaptable for policy changes (e.g., GDPR, IIROC, CSA).

Document all compliance measures for audit readiness.

vii. Market Acceptance and Adoption

Mitigation Strategy:

Launch awareness campaigns and educational content to reduce user resistance. Partner with influencers and financial advisors for outreach. Offer limited-time free trials and interactive onboarding experiences to build trust and familiarity.

viii. Competitive Landscape

Mitigation Strategy:

Continuously evaluate competitors and update the feature roadmap to maintain differentiation. Emphasize the app's unique HQM analysis, educational modules, and user-focused design. Prioritize iterative development based on user feedback and market trends.

ix. Economic Conditions and Market Volatility

Mitigation Strategy:

Regularly retrain recommendation models to reflect changing market dynamics. Offer users risk-averse investment options and simulations. Provide in-app resources on navigating economic uncertainty to maintain engagement during downturns.

4.2.2 Contingency Plans for Addressing Unforeseen Challenges

To ensure resilience against unexpected disruptions, the following contingency plans have been defined for critical risk scenarios:

i. System Downtime or Application Failure

Contingency Plan:

Deploy a dedicated incident response team with a pre-defined recovery protocol. Maintain real-time communication with users through alerts and status updates. Offer compensation where necessary (e.g., service credits or free extensions) to maintain user goodwill.

ii. Regulatory Changes

Contingency Plan:

Retain access to legal and compliance consultants to evaluate and interpret emerging regulations.

Design modular and adaptive system components to quickly accommodate new legal requirements without significant refactoring.

iii. Low Market Adoption

Contingency Plan:

Conduct in-depth user feedback sessions and behavioural analytics to uncover pain points. Revise the marketing strategy and product positioning. Prioritise the development of high-impact features based on user expectations. Consider partnerships or integration with more familiar platforms to increase visibility and credibility.

4.3 Challenging Component: Top 3 Risks and Alternative Strategies

This section outlines the top three most critical risks, along with a multi-layered approach comprising primary, backup, contingency, and alternative strategies for each.

i. System Downtime or Application Failure

Primary Strategy: Perform regular maintenance, maintain automated backups, enforce infrastructure redundancy, and enable 24/7 application performance monitoring.

Backup Strategy: Adopt a rolling deployment model to minimize disruption during updates. Use feature flagging to enable controlled feature rollouts and minimize the risk of full-system failures.

Contingency Plan: Activate the incident response team immediately upon failure. Notify users through in-app notifications and email alerts. Provide compensatory offerings like credits or trial extensions to maintain satisfaction.

Alternative Strategy: Introduce edge computing to decentralize critical components and ensure basic functionality in the event of central server failure. Utilize distributed systems for fault tolerance and high availability.

ii. Inaccurate Stock Recommendations Due to Data Errors

Primary Strategy: Source financial data from multiple trusted providers. Integrate automated data validation and cleansing processes to minimize data-related issues.

Backup Strategy: Implement redundancy through parallel data pipelines and real-time monitoring systems that flag inconsistencies or lag.

Contingency Plan: Include a system override mechanism that pauses new stock recommendations in the event of severe data corruption or failure, alerting both users and internal teams for resolution.

Alternative Strategy: Design fallback algorithms capable of generating basic recommendations using cached or partial data. This ensures continuity of service even during data outages.

iii. Market Acceptance and Adoption

Primary Strategy: Launch targeted marketing campaigns, leverage influencer partnerships, and provide engaging educational content to build trust among retail investors.

Backup Strategy: Initiate user referral and rewards programs. Foster organic growth through social proof and testimonials. Engage directly in investment communities and forums.

Contingency Plan: Redesign the onboarding experience and user interface to reduce friction. Launch pilot programs and gather feedback for product realignment. Explore bundling features with partner platforms or fintech institutions.

Alternative Strategy: Introduce co-branded versions of the app in collaboration with well-known financial service providers. Use their reputation to access a broader and more trusting user base.

PROJECT BUDGET REPORT

5.1 Introduction

The project budget is a critical component of the overall project planning process, serving as a financial blueprint for the successful development, deployment, and maintenance of the Stock Selection and Financial Literacy App. This chapter outlines the estimated costs, justifications, allocations, and contingency planning necessary to ensure smooth execution and delivery within scope, time, and quality constraints.

5.2 Budgeting Objectives

The objectives of this budgeting process include:

- Estimating the total cost of development, testing, marketing, deployment, and ongoing support
- Identifying and allocating funds to major cost categories and subcategories
- Ensuring optimal resource utilization while maintaining high-quality standards
- Planning for contingencies and unexpected events to mitigate financial risks
- Aligning the budget with the project timeline and deliverables over a 6–8 month duration

5.3 Total Budget Overview

Category	Estimated Cost (CAD)	Percentage of Total
1. Development	\$25,000	50%
2. Testing	\$5,000	10%

3. Marketing	\$7,000	14%
4. Ongoing Maintenance	\$5,000	10%
5. Deployment & Support	\$8,000	16%
Total	\$50,000	100%

5.4 Detailed Budget Allocation

This section breaks down the total project budget into specific categories and subcategories, providing a granular view of how funds will be utilized. Each category represents a critical aspect of the software development lifecycle and contributes directly to the successful execution of the project.

5.4.1 Development Budget (\$25,000)

Development costs include all activities related to designing and building the application's core functionality. This encompasses both the frontend and backend systems, the database architecture, and third-party API integrations.

Subcategory	Estimated Cost (CAD)	Description
Frontend Development	\$7,500	Dashboard UI, mobile interface using React and Flutter
Backend Development	\$8,000	Core logic, HQM analysis, user management system
Database Design	\$4,500	Schema design, cloud storage with Firebase or AWS RDS
API Integration	\$5,000	Real-time data from financial APIs like IEX Cloud, Alpha Vantage

5.4.2 Testing Budget (\$5,000)

Testing ensures the stability, security, and usability of the application before it reaches end users. It includes manual, automated, and user acceptance testing to validate all components of the system.

Subcategory	Estimated Cost (CAD)	Description
Manual Testing (QA)	\$2,000	Functional tests, bug tracking, and manual validation of workflows
Automated Testing	\$2,000	Regression testing scripts to ensure new features do not break existing ones
User Acceptance Testing (UAT)	\$1,000	Stakeholder testing to ensure the application meets business needs

5.4.3 Marketing Budget (\$7,000)

Marketing is essential for attracting users and establishing the application's presence in the financial technology space. It involves branding, advertising, and outreach campaigns designed to increase visibility and adoption.

Subcategory	Estimated Cost (CAD)	Description
Branding	\$2,000	Logo, UI assets, and brand identity development
Paid Advertising	\$3,000	Google Ads, Facebook, and Instagram sponsored posts
User Campaigns	\$2,000	Referral programs, promotional webinars, influencer partnerships

5.4.4 Ongoing Maintenance Budget (\$5,000 Annually)

Ongoing maintenance ensures that the application remains functional, secure, and relevant after deployment. This budget covers bug fixes, new feature implementation, and customer support services.

Subcategory	Estimated Cost (CAD)	Description
Bug Fixes	\$2,000	Addressing reported issues, errors, and system crashes
Feature Updates	\$2,000	Enhancing app functionality based on user feedback
User Support	\$1,000	Providing technical support and maintaining help resources

5.4.5 Deployment & Customer Support Budget (\$8,000)

This budget supports the final launch of the application, including hosting setup, CI/CD pipeline integration, and the provision of live support services for users during the onboarding process.

Subcategory	Estimated Cost (CAD)	Description
Cloud Infrastructure Setup	\$3,000	Hosting configuration, domain setup, CDN services

Deployment Execution	\$2,000	CI/CD pipeline configuration and live server launch
Customer Service	\$3,000	Initial user support and feedback handling

5.5 Resource Costing Breakdown

This section outlines the cost of human resources, technology tools, and external services that are required for successful project execution. These estimates are based on standard industry rates and usage forecasts.

Resource Type	Role / Item	Rate / Unit Cost	Estimated Cost
Human Resources	2 Frontend Developers	\$50/hr × 160 hrs	\$16,000
Human Resources	2 Backend Developers	\$55/hr × 160 hrs	\$17,600
Human Resources	QA Tester	\$40/hr × 100 hrs	\$4,000
Human Resources	Project Manager	\$60/hr × 80 hrs	\$4,800
Technology	Firebase/AWS	\$100/month × 12	\$1,200
Technology	Paid APIs	\$200/month × 12	\$2,400
Technology	Dev Tools & Licenses	One-time	\$1,000
External Services	UI/UX Consultant	\$75/hr × 30 hrs	\$2,250
External Services	Legal Compliance Review	One-time	\$2,500
External Services	Marketing Consultant	One-time	\$3,000

5.6 Contingency Budget (\$6,000)

To ensure financial preparedness for unforeseen events, a contingency budget of \$6,000 (10% of the total project budget) is allocated. This reserve fund will be used for unplanned scenarios such as critical bug fixes, API service disruptions, or compliance changes.

Typical use cases include:

- Emergency server upgrades or migrations
- Expedited security patches or compliance updates
- Temporary hiring of freelance developers
- Backup services and disaster recovery mechanisms

5.7 Cost Summary

Category	Amount (CAD)
Development	\$25,000
Testing	\$5,000
Marketing	\$7,000
Ongoing Maintenance	\$5,000
Deployment & Support	\$8,000
Contingency	\$6,000
Total Estimated Budget	\$56,000

5.8 Budget Alignment with Project Timeline

The budget is strategically aligned with the project Work Breakdown Structure (WBS) and the Gantt Chart outlined in Chapter 3. Major expenses occur during the Design, Development, and Deployment phases, ensuring that all key activities are fully supported by appropriate financial planning.

- Planning Phase: Legal review, PM resource setup
- Design Phase: UI/UX consulting, tools, wireframes
- Development Phase: Major development costs, backend & frontend
- Testing Phase: QA and UAT processes
- Deployment Phase: Hosting setup, marketing, support

5.9 Conclusion

This detailed budget report reflects the comprehensive financial strategy necessary to deliver a high-quality Stock Selection and Financial Literacy App. The allocations are realistic, well-reasoned, and

tied directly to project objectives and timelines. With provisions for human resources, marketing, risk mitigation, and long-term maintenance, this financial plan positions the project for timely and successful delivery with room to adapt to future needs.

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